# Jitesh **Jain**

**Ph.D. Student | Computer Science | Georgia Tech**

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## Education

**Georgia Tech** — *Atlanta (GA)*

Ph.D. Student in Computer Science — *Since Fall 2023; Expected Spring 2027*

- Advisor: Prof. Humphrey Shi
- CGPA: 4.0/4.0

**Indian Institute of Technology Roorkee** — *Roorkee, India*

B.Tech. in Computer Science and Engineering — *2019 - 2023*

- CGPA: 9.0/10

## Experience

**NVIDIA** — *Santa Clara (CA)*

High Performance AI Intern w Prof. Humphrey Shi — *Jan 2026 - Present*

- *Research:* Training Coding Agents for CUDA Kernel Optimization.

**PRIOR @ Allen AI** — *Seattle (WA)*

Research Intern w/ Dr. Christopher Clark — *May 2025 - Dec 2025*

- *Research:* Training Long Video Understanding Agents with Reinforcement Learning.

**Microsoft Research** — *Redmond (WA)*

Research Intern w/ Dr. Jianwei Yang — *May 2024 - Dec 2024*

- *Research:* Improving Visual Perception in MLLMs with Embedding Distillation.

**SHI Labs @ UIUC & U of Oregon** — *Eugene (OR)*

Research Intern w/ Prof. Humphrey Shi — *November 2020 - June 2023*

- *Research:* Unifying Image Segmentation with a Single Architecture, Model and Dataset.

## Publications

For a complete list of my research works, please lookup my **google-scholar**.

**SAGE: Training Smart Any-Horizon Agents for Long Video Reasoning with Reinforcement Learning** — *CVPR 2026*

**Jitesh Jain**, Jialuo Li, Zixian Ma, Jieyu Zhang, Chris Dongjoo Kim, Sangho Lee, Rohun Tripathi, Tanmay Gupta, Christopher Clark, Humphrey Shi — *PRIOR @Allen AI*

- **[Paper] [GitHub] [Project Page]**

**Elevating Visual Perception in Multimodal LLMs with Visual Embedding Distillation** — *NeurIPS 2025*

**Jitesh Jain**, Zhengyuan Yang, Humphrey Shi, Jianfeng Gao, Jianwei Yang — *Microsoft Research, Redmond*

- **[Paper] [GitHub] [Project Page]**

**AUGUSTUS: An LLM-Driven Multimodal Agent System with Contextualized User Memory** — *LAW @NeurIPS 2025*

**Jitesh Jain***, Shubham Maheshwari*, Ning Yu, Wen-mei Hwu, Humphrey Shi — *SHI Labs @ GaTech*

- **[Paper] [GitHub][Project Page]**

**VCoder: Versatile Vision Encoders for Multimodal Large Language Models** — *CVPR 2024*

**Jitesh Jain**, Jianwei Yang, Humphrey Shi — *SHI Labs @ GaTech*

- **[Paper] [GitHub] [Project Page]**

**OneFormer: One Transformer to Rule Universal Image Segmentation** — *CVPR 2023*

**Jitesh Jain**, Jiachen Li*, MangTik Chiu*, Ali Hassani, Nikita Orlov, Humphrey Shi — *SHI Labs @ UIUC & UO*

- **[Paper] [GitHub] [Project Page]**

**Keys to Better Image Inpainting: Structure and Texture Go Hand in Hand** — *WACV 2023*

**Jitesh Jain***, Yuqian Zhou*, Ning Yu, Humphrey Shi — *SHI Labs @ UIUC & UO*

**[Paper] [GitHub] [Project Page]**

## Honours & Achievements

2026 **Selected for CVPR 2026 Doctoral Consortium**, A competitive program pairing promising PhD researchers with senior computer vision mentors for research and career guidance at CVPR 2026.

2025 **NeurIPS Scholar Award**, Awarded a full conference registration and a 7-night San Diego hotel stay to attend NeurIPS 2025.

2022 **Bronze Medalist**, Awarded the Bronze Medal under Bosch's Age and Gender Detection Track at Inter IIT Tech Meet 10.0. The competition is held every year among all the IITs.

2019 **All India Rank - 252**, Joint Entrance Exam (JEE) Advanced is attempted by more than 230,000 students across India for admission into the IITs.

2019 **All India Rank - 454**, Joint Entrance Exam (JEE) Main is attempted by more than 1,000,000 students across India to gain eligibility for attempting JEE Advanced.

2018 **KVPY Fellowship Holder**, KVPY is a highly prestigious National Fellowship awarded by the Indian Institute of Science to students showing high aptitude in research.

2017 **NTSE Scholarship Awardee**, NTSE is a national level examination for the secondary school level to identify and recognize students with high intellect and skills.

## References

**Dr. Humphrey Shi** *United States*

Associate Professor @Georgia Tech | Vice President @NVIDIA

- **Email**: shi@gatech.edu

**Dr. Judy Hoffman** *United States*

Associate Professor @University of California, Irvine

- **Email**: judy.hoffman@uci.edu

**Dr. Jianwei Yang** *United States*

Member of Technical Staff @xAI

- **Email**: jw2.yang@gmail.com

**Dr. Christopher Clark** *United States*

Research Scientist @Allen AI

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**Dr. Ning Yu** *United States*

Lead Research Scientist @Netflix Eyeline Studios

- **Email**: ningyu.hust@gmail.com